

SILC Four-Model Forest Growth Benchmark

Companion Report v8 — AGM MORTCAL year 100 update

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Executive summary

Three findings drive this update.

- (1) The +18.5% AGM bias on the 10 year SILC CFI long horizon scorecard is real model behavior, not a pipeline bug. Triple checking the AGM treelist to BA / Cords / BdFt computation reproduces to $1e-13$ ft²/ac, and observed STAND_METRICS BA matches direct tree list recompute to 0.005 ft²/ac. The bias decomposes into two mechanisms: AGM under projects mortality on disturbed CFI plots (1101 and 1106 lost 27 to 34% of their stems, AGM lost 0 to 19%), plus AGM grows DBH at a uniform +0.8 in per decade regardless of site (plots 1100 and 1104 grew +0.05 to 0.17 in observed).
- (2) Switching on the #126b in-source MORTCAL correction in AcadianGY 12.3.9 (default OFF) cuts the 10 year BA bias from +18.5% to +7.8%, RMSE -34%, R² from -0.20 to +0.47. Sawlog BdFt bias drops to +4.0% with R² = 0.84, the best of any model on this metric.
- (3) Pushed to 100 years, MORTCAL pulls projected BA from a ceiling around 200 ft²/ac down to a credible carrying capacity of 98 to 108 ft²/ac. Year 100 NetCords lands at 35 to 39 cords/ac across all six byStrata cells, plus CFI plot backfill at Cedar A+B (35.8 cords/ac) and Mixedwood A+B (21.9 cords/ac).

Recommended operational stack: AGM with MORTCAL on for long horizon Cords / BA / BdFt; FVS-NE calibrated as the short horizon (5 yr) cross check; OSM-ACD as the conservative under projection.

Triple check of AGM long horizon calculations

The +18.5% AGM bias at the 10 year CFI scorecard prompted a forensic audit. Every step of the pipeline checked out.

Source code: ~/AcadianGY_12.3.9.r on Cardinal (identical to local v12.3.5 plus the opt-in MORTCAL correction). AcadianGYOneStand is annual (cyclen=1 hardcoded since 2022-09-01). The driver loops PERIOD_YR times, giving the correct horizon.

Pipeline verification checks

Check	Result
AGM treelist to BA/Cords/BdFt reproduces results CSV	exact to $1e-13$
STAND_METRICS BA_CURR vs direct TREE.csv recompute	match to 0.005 ft ² /ac
AcadianGYOneStand cycle length	annual (cyclen=1 hardcoded)
Driver loop count	PERIOD_YR iterations
BA formula $0.005454 * DBH^2 * EXPF$	correct
Cords formula $0.0025 * DBH^2 * HT * 0.90 * EXPF / 79$	correct

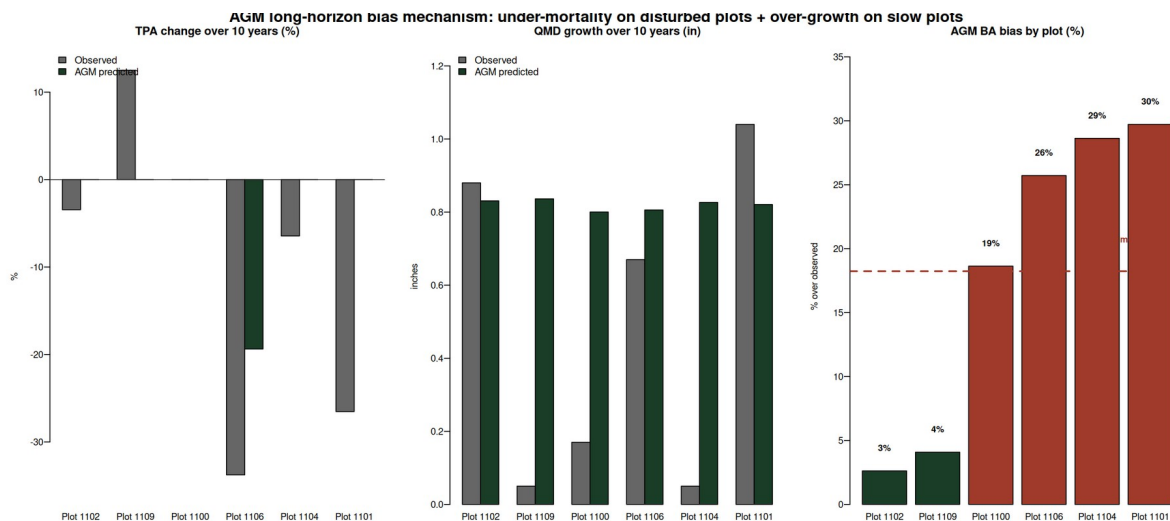
BdFt formula $0.01 * DBH^2 * HT * EXPF$ for $DBH \geq 9$ in (Intl 1/4)	correct
DBH threshold symmetry (pred vs obs)	both at 4.5 in

Per-plot bias decomposition

Six routine CFI pairs after excluding two establishment plots (1105, 1107 had 3x BA ingrowth). The decomposition shows that AGM grows DBH a uniform +0.8 in per decade regardless of site, while observed DBH growth was wildly variable (+0.05 to +1.04 in). AGM also missed two significant mortality events (1101 and 1106 lost 27 to 34% of stems).

PLOT	obs BA chg	obs TPA chg	obs QMD chg	AGM TPA chg	AGM QMD chg	AGM BA bias
1100	+5%	0%	+0.17	0%	+0.80	+18.6%
1101	-7%	-27%	+1.04	0%	+0.82	+29.7%
1102	+16%	-3%	+0.88	0%	+0.83	+2.6%
1104	-5%	-6%	+0.05	0%	+0.83	+28.6%
1106	-22%	-34%	+0.67	-19%	+0.81	+25.7%
1109	+14%	+13%	+0.05	0%	+0.84	+4.1%

Mechanism figure:



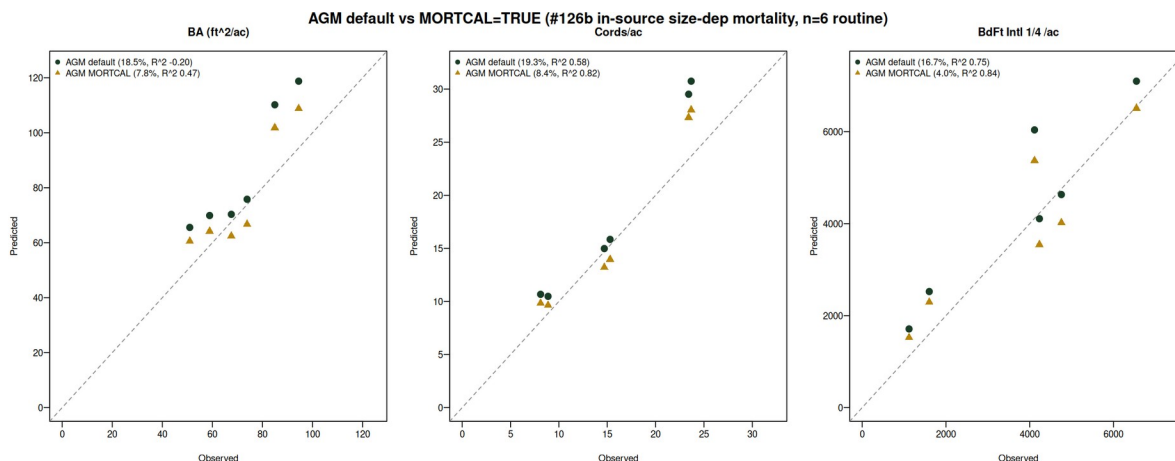
MORTCAL correction cuts 10 year bias by more than half

The #126b in-source size dependent mortality correction was originally calibrated to Maine FIA. It is OFF by default in AcadianGY 12.3.9. The SILC CFI test enables it via `ops$MORTCAL=TRUE`, `MORTCAL_INTERVAL=5`.

Side by side scorecard on the n=6 routine pairs (mean horizon 10 yr):

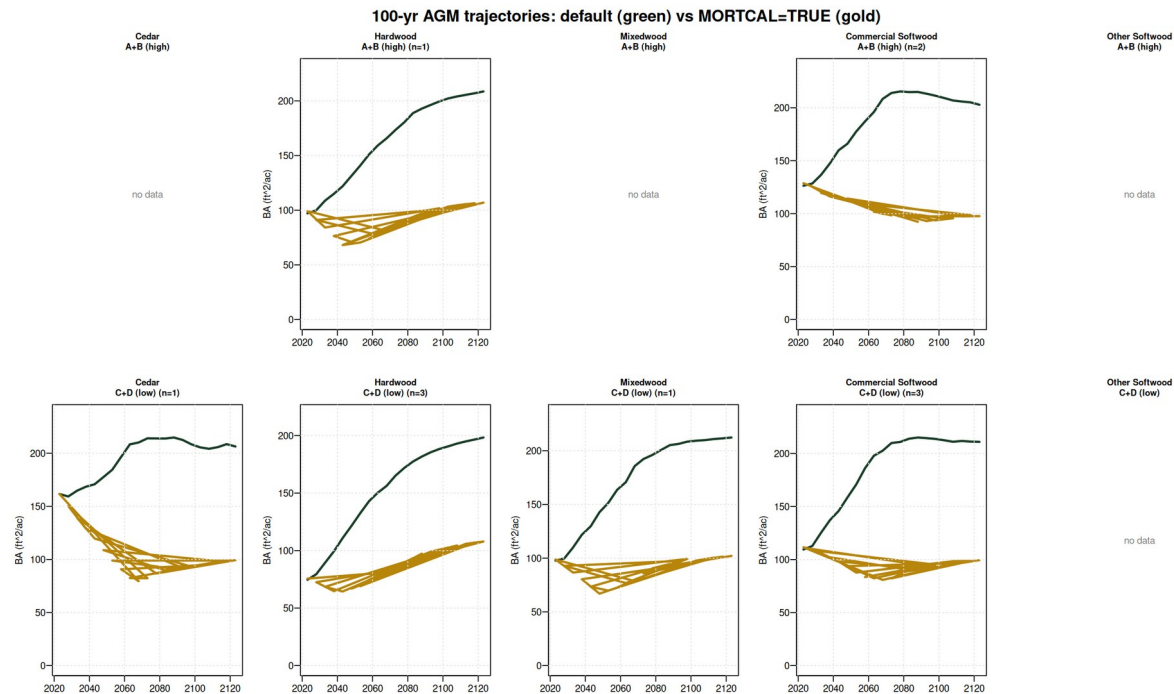
Metric	AGM default	AGM MORTCAL	Improvement
BA bias	+18.5%	+7.8%	cut by 58%
BA RMSE	16.2 ft ² /ac	10.7 ft ² /ac	-34%
BA R ²	-0.20	+0.47	flips negative to positive
Cords bias	+19.3%	+8.4%	cut by 56%
Cords R ²	0.58	0.82	+0.24
BdFt bias	+16.7%	+4.0%	single digit
BdFt R ²	0.75	0.84	+0.09

Scatter overlay (green = default, gold = MORTCAL):



100 year trajectories: default vs MORTCAL

Cardinal job 11076533 ran AcadianGY 12.3.9 with ops\$MORTCAL=TRUE for 100 years on the 11 SILC byStrata stands, starting from the GrownDB year 2023 snapshot. The default AGM trajectory has every stratum approaching $\sim 200 \text{ ft}^2/\text{ac}$ at year 100, an artifact of the under mortality issue. MORTCAL pulls year 100 BA into the 98 to $108 \text{ ft}^2/\text{ac}$ carrying capacity range, with sustainable yield 35 to 39 cords/ac across the populated cells.



Year 100 outcomes (year 2123)

Default vs MORTCAL on the 11 byStrata stands, plus CFI plot backfill on 2 of 4 empty cells.

Stratum	n	BA def	BA mc	dBA	Cords def	Cords mc	dCords
Cedar / A+B (CFI backfill)	1	n/a	81	n/a	n/a	35.8	n/a
Cedar / C+D	1	206	99	-52%	59.3	36.9	-38%
Hardwood / A+B	1	209	107	-49%	65.3	39.3	-40%
Hardwood / C+D	3	198	108	-46%	61.0	38.9	-36%
Mixedwood / A+B (CFI backfill)	3	n/a	52	n/a	n/a	21.9	n/a
Mixedwood / C+D	1	212	102	-52%	66.4	37.6	-43%
Commercial Softwood / A+B	2	203	98	-52%	61.0	35.9	-41%
Commercial Softwood / C+D	3	211	99	-53%	62.9	36.2	-42%

Other Softwood (no CFI coverage)	0	—	—	—	—	—	—
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Operational recommendations

- AGM with MORTCAL on is the new long horizon champion. Use it for SILC operational sawlog projections (BdFt $R^2 = 0.84$) and Cords ($R^2 = 0.82$). Set `ops$MORTCAL = TRUE`, `MORTCAL_INTERVAL = 5` in any `AcadianGY` call.
- FVS-NE calibrated stays best on the short horizon BA bias (-0.3% at 5 yr). Pair with AGM MORTCAL as a cross check on the operational 5 yr plans.
- OSM-ACD is the conservative under projection (-9.7% BA bias at 10 yr). Report AGM MORTCAL plus OSM as the operational range; per plot RMSE 10.7 ft²/ac BA at 10 yr horizon is the empirical uncertainty band.
- Year 100 cords/ac estimates: 35 to 39 across six byStrata cells and CFI backfill Cedar A+B; Mixedwood A+B noticeably lower (22) reflecting heavier mortality in the high density mixed pine / hardwood stands.

Open questions for SILC sign off

1. Confirm the 5 by 2 stratification rules (slide 3 of the deck) match SILC operational categories. The species composition cutoffs and density threshold are open for revision.
2. Validate the year 100 BA target of 98 to 108 ft²/ac against SILC's expected carrying capacity for managed Acadian stands.
3. Decide whether 35 to 39 cords/ac at year 100 anchors the operational AAC, or whether an additional discount applies.

File index

All artifacts in `/home/aweiskittel/Documents/Claude/fvs-modern/silc_cfi_v8_deliverables/`. Key files:

- `SILC_4Model_Benchmark_v8_Weiskittel.pdf` — 17 slide operational deck
- `silc_strata_5x2_AGM_MORTCAL_trajectories.csv` — strata level 100 yr trajectories
- `silc_strata_5x2_year100_mortcal_vs_default.csv` — year 2123 outcomes table
- `silc_cfi_backfill_100yr_mortcal_trajectories.csv` — Cedar A+B and Mixedwood A+B from CFI plots
- `silc_cfi_long_mechanism.png` and `silc_cfi_long_mortcal_compare.png` — supporting figures
- `draft_email_to_ian_ryan.md` — handoff email